JULIAN ESTEBAN RINCÓN RODRÍGUEZ

Computer Science and Artificial Intelligence | Data Engineering · Machine Learning · Backend
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PROFESSIONAL SUMMARY

Final-year (9th of 9 semesters) Computer Science and Artificial Intelligence student focused on data engineering, machine learning, and backend development. I build end-to-end data pipelines on AWS, machine learning models validated on held-out data with confidence intervals, and tested Python services that run every day on Linux. Experience delivering software to clients and working in teams of three to five. Seeking a six-month data or software engineering internship for January – June 2027.

EDUCATION

Universidad Sergio Arboleda — Bogotá, Colombia | 2022 – Present
B.Sc. Computer Science and Artificial Intelligence | 9th of 9 semesters | GPA 4.04/5.0, with the last three semesters above 4.5/5.0.

Relevant coursework: Big Data and Data Engineering, Machine Learning, Neural Networks, Natural Language Processing, Modeling and Simulation.

PROFESSIONAL EXPERIENCE

Full-Stack and AI Developer (Freelance) — Zafra Asesores Tributarios | Girardot, Colombia | Jun. 2026

- Built a sales CRM for a tax advisory firm, modeling clients, prospects, campaigns, and interactions in a FastAPI/SQLAlchemy backend with a React/Vite view of the prospect pipeline.
- Added an LLM layer (Groq, Llama 3.3 70B) that summarizes interactions, drafts emails, and scores leads, plus a lead-qualification bot on the WhatsApp Cloud API.
- Demoed and handed the system over with technical documentation and a scoped plan (authentication and continuous deployment) for the production rollout.

Full-Stack Developer — ReushiGo | Bogotá, Colombia | May 2026
Ordering platform for a family coffee distribution business · reushi-go.vercel.app

- Built the admin panel (React/Vite/Tailwind) and the Express/Prisma backend, protected with admin-key authentication and rate limiting.
- Implemented a Telegram sales bot that takes orders in natural language and through interactive menus over a live 19-product catalog.
- Set up CI on GitHub Actions (lint, build, npm audit, tests) and documented a security review of the project.

SELECTED PROJECTS

SAVI — Autonomous Real Estate Valuation Agent | 2026 | v1–v2 team of 4, v3 individual
Python · XGBoost · PyTorch (Double DQN) · AWS Lambda · EC2 · S3 · SageMaker · RAG · Claude API

- Audited the team version and found that 93% of its dataset was assessor records padded with constant columns; rebuilt the data from public sources (2,930 sales joined to 18,078 parcels, with FHFA and Zillow indices).
- Reached out-of-fold R^2(log) = 0.921 and 7.8% MAPE with XGBoost, and trained RL policies (Value Iteration, Q-Learning, Double DQN) that approve, review, or reject valuations, tested once with 95% bootstrap intervals.
- Re-engineered it alone as an event-driven AWS pipeline (S3 → Lambda → EC2 → Lambda → SageMaker Spot) with boto3 infrastructure as code and automatic shutdown, at under US$0.05 per run; 191 tests and CI on GitHub Actions.
- Deployed a RAG agent with 8 tools (Claude tool-use loop with a deterministic fallback) behind a public web interface on a Lambda Function URL.

ShopStream — AWS Big Data Pipeline | May – Jul. 2026
PySpark · AWS Lambda · EMR · Glue · RDS PostgreSQL · Flask · Zappa

- Built an end-to-end pipeline for 2.5M e-commerce events: partitioned S3 ingestion, Lambda validation with CloudWatch metrics and quarantine, EMR/PySpark analytics for six KPIs and anomaly detection, Glue ETL, and an RDS PostgreSQL warehouse.
- Reached 86% test coverage with pytest, automated CI/CD with GitHub Actions, and locked down RDS security groups.

AWS Big Data Platform (Chinook / NYC Taxi) | Mar. – May 2026 | Team of 3
EC2 · RDS · Athena · S3 · Glue · Terraform · FastAPI · React

- Processed the NYC Yellow Taxi dataset (94M rows, 12 months) on EMR with PySpark, fixing schema drift across Parquet files with an explicit StructType and unionByName normalization.
- Deployed a full-stack app (React/Vite and FastAPI on two EC2 instances, PostgreSQL on RDS) and a Hive-partitioned Athena table over S3.

Phantom — Local Multi-Agent Coding Workspace | Sep. 2026 – Present
Rust · TypeScript · Next.js · Tauri · MCP · Whisper · systemd · Tailscale

- Extended a fork of the open-source Codeg project (Apache-2.0) into one workspace for Claude Code, OpenCode, and Hermes Agent, with a shared timeline, agent-to-agent delegation, and a fully local voice layer (Whisper large-v3-turbo on CUDA).
- Built a delegation router that picks agent, model, and effort per task, ranks models by measured success (95% Wilson interval), and switches to a cheaper profile near quota limits; it runs as loopback-only systemd services reachable from a phone over Tailscale, with a 12-check self-verifying deploy.

Ayuda Terremoto Colombia — Distributed Humanitarian Coordination | Aug. 2026 – Present
FastAPI · PostgreSQL · React · Vite · IndexedDB · Render

- Built an offline-first coordination platform after the magnitude 7.4 Chocó earthquake: a FastAPI/PostgreSQL central node and a React/IndexedDB local node with a per-action retry sync queue.
- Implemented fuzzy-match report deduplication, automatic activation from USGS polling, and HXL export, covered by 80+ pytest tests.

NEXUS — Personal AI Assistant | 2025 – Present (v2 since Oct. 2026)
Python · MCP · SQLite FTS5 · systemd · Whisper · Chatterbox · Tailscale · GCP

- Re-architected a ~47,000-line v1 into a ~8,800-line Python layer on Phantom with 1,400+ tests and no root access: daily briefing, an agenda that turns emails into Google Calendar events, push-to-talk voice, and memory shared by three AI agents over MCP.
- Built a copilot that repairs test-breaking commits in an isolated git worktree and merges only after review, a Telegram gym log with deterministic load progression, and a cloud-VM watchdog that alerts on outages.

Additional projects: Victor (Jul. 2026 – Present), a voice race engineer for Linux sim racing (C shared-memory reader, LLM provider failover, 219 tests, CI on Python 3.10 and 3.12); a Sentiment140 lab tracked in MLflow (team of 5, champion model at 0.824 macro-F1 served by FastAPI on EC2); Dino Island, a generative NLP pipeline (team of 5: character-level GRU, gemma4 via Ollama on SageMaker, Lambda app); an NLP API with spaCy and FastAPI (93/100 in a live evaluation); a programming language built with ANTLR4; RL and CNN labs.

TECHNICAL SKILLS

Languages: Python (intermediate-advanced), SQL, C, Bash; TypeScript, Rust, and JavaScript (basic)

Machine Learning and AI: scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, spaCy, MLflow, Whisper, Ollama

Big Data and Cloud: AWS (EC2, S3, RDS, Athena, Glue, EMR, Lambda, SageMaker), GCP, PySpark, Terraform, Docker (basic)

Data and ETL: Pandas, NumPy, Matplotlib, Power BI, Jupyter Notebook, openpyxl

Backend and Web: FastAPI, Flask, Express, React, SQLAlchemy, Prisma, PostgreSQL

Linux and Tooling: Linux (Fedora as daily OS, systemd, cgroups), Git, GitHub Actions, pytest, ChromaDB, MCP, LLM APIs (Claude, Groq, Cerebras, NVIDIA NIM, Gemini)

ACHIEVEMENTS

1st Place — Wisibilízalas International Competition, Universitat Pompeu Fabra | 2019

- Led the winning web application as technical product manager and frontend/UX lead, representing Ecuador in the Senior category at 17 against teams from Latin America and Europe.

1st Place — USABOT Robotics Challenge | November 2025

- Directed the CAD design and sensor architecture of a smart automated irrigation system and delivered the commercial pitch, winning against university teams.

CERTIFICATIONS

AWS Academy Graduate — Data Engineering | Amazon Web Services Academy | May 2026 (40 hours)

Database Programming with SQL | Oracle Academy | December 2024

Database Design | Oracle Academy | August 2024

LANGUAGES

Spanish: native. English: certified B2 overall (iTEP Academic-Plus, November 2025; Listening C1, Reading and Speaking B2).